

LLMs for corpus annotation: Close and extended replications of Morin & Marttinen Larsson (2025) using smaller open LLMs

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Abstract

Large language models (LLMs) are increasingly being leveraged for corpus annotation. In an early evaluation of such an application, *Linguistics Vanguard* published “a replicable method for automating grammatical annotation” (Morin & Marttinen Larsson 2025). Using Claude 3.5 Sonnet, it involved several ‘conversational’ phases of training and testing. The study reported an accuracy of 93% and concluded that the pipeline may also be effective for other linguistic tasks, including in languages other than English. We first aimed to replicate this “replicable method” as closely as possible, using the same data, but with two open LLMs. While a 24B Mistral model did not perform as well (67% accuracy), a 120B open GPT model yielded equally good results (94% accuracy). This suggests that, as of 2026, large commercial LLMs may not be necessary for this type of annotation task. In a second replication study, we tested the need for the extensive ‘in-conversation’ training process outlined in Morin & Marttinen Larsson (2025). We compared it to a much simpler few-shot training approach, concluding that the training process need not be as elaborate as in the original study. Our results have important implications for the replicability of linguistic research involving LLMs. Although the authors published their data and prompts, we found that crucial information necessary to replicate the procedure was missing. We argue that issues around accessibility, reproducibility, data leakage, and carbon footprint can and should be taken into consideration when choosing a tool and warn against the risk of overgeneralising the results of a single, binary classification task in English to other linguistic tasks and languages.

Keywords: Replication, Large language model, Reproducibility, Corpus linguistics, Open Science.

1 Introduction

Over the past decades, linguists have gained access to ever larger corpora of natural language. This poses great opportunities, but also greatly increases the workload associated with the manual and/or (semi-)automated preprocessing and annotation of large amounts of data. The rise of large language models (LLMs) is therefore an exciting development for linguistics research, offering new possibilities for complementing and extending existing automated methods for processing larger datasets and corpora. In recent years, a growing number of publications have reported on the usefulness of LLMs for a range of linguistic annotation tasks, such as the annotation of speech acts (Yu et al. 2024; Su & Ye 2025; Ostyakova, Mikhailova & Kononov 2025), metaphor detection (Fuoli et al. 2026), word sense or function disambiguation (Fonteyn et al. 2024; Morin & Marttinen Larsson 2025), and political text classification and annotation (Heseltine & Clemm Von Hohenberg 2024).

While LLMs are increasingly prevalent in both linguistic research and our everyday lives, they also raise a number of concerns. For research purposes, specifically, their ‘black-box’ nature raises concerns over the transparency, reproducibility, replicability of the research produced with their assistance. Note, that although the terms have been used interchangeably in the past, we distinguish between *reproducibility* and *replicability* following the definitions of the FORRT (Parsons et al. 2022) and Turing Way (Community 2022) communities. We thus define *reproducibility* as the ability of an independent researcher or team to obtain the same results as an original study using the original data and methods. This is in contrast to *replicability*, where the same methods are applied, but on different data; and robustness, where the same data, but different methods are used (see Figure 1). Only if an effect can be reliably observed across different datasets with different methods, can we claim that it is *generalisable*.

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Figure 1: Term disambiguation (Community 2022, CC BY 4.0)

In this study, we seek to replicate Morin & Marttinen Larsson (2025)’s methodological study, whose title claims to present a “replicable method” for leveraging LLMs for linguistic annotation. The original study was conducted using a commercial, closed-source LLM: Anthropic’s model Claude 3.5 Sonnet (Morin & Marttinen Larsson 2025: 4). It is no longer possible to reproduce

the original results as Claude Sonnet 3.5, which was released in June 2024, was retired in October 2025 (<https://web.archive.org/web/20260911184540/https://killedbyai.net/dead/claude-3-5-sonnet/>), just 18 months after its release and only six months after the online version of the original article was published. We therefore sought to apply the original methodological pipeline to the same data, but using two open-source LLMs, whose accessibility can be guaranteed for independent verification and future replications and whose parameters can be controlled to generate reproducible results (see Section 3). In an extended replication, we subsequently re-ran the classification task with a much simpler pipeline than that proposed in the original paper (see **?@sec-ExtRep**).

In the meantime, the authors of the original study have published a follow-up paper as a preprint (Morin & Larsson 2026), in which they further develop their “large-scale pipeline for automatic corpus annotation using LLMs” based on the same case study of the English *consider* construction. Although there are many similarities in its aims, this follow-up study adopts considerably simpler prompting than Morin & Marttinen Larsson (2025) and the case study involves a larger dataset. It also pursues a somewhat different focus, placing greater emphasis on the different types of evaluative uses of the *consider* construction and on the role of genre. However, the classification tasks are again conducted using a large, commercial model – this time from OpenAI (Morin & Larsson 2026: 17). As such, we believe that evaluating the reproducibility of the original pipeline with smaller, open LLMs remains a very relevant and worthwhile undertaking.

While large, cloud-based, proprietary LLMs continue to top performance benchmarks, open-source LLMs have made significant progress in closing this performance gap (Manchanda et al. 2025: 4). We consider open models to form a crucial part of the open research ecosystem (alongside open materials, open data, open code, open documentation, open access, etc.). Openness promotes transparency, reproducibility, and quality control; which are necessary features to support accessible, reliable, and credible empirical findings (Liesenfeld, Lopez & Dingemanse 2023). In the discussion and conclusion section of their subsequent study, Morin & Larsson (2026) detail the limitations of using on closed-source, commercial LLMs; however, both their studies do not test the necessity of relying on these systems to achieve high accuracy. This is what we set out to do in the present study.

2 Research aims and questions

To begin, the present study aims to replicate the results published in:

Morin, Cameron, and Matti Marttinen Larsson. 2025. “Large Corpora and Large Language Models: A Replicable Method for Automating Grammatical Annotation.” *Linguistics Vanguard* 11 (1): 501–10. <https://doi/10.1515/lingvan-2024-0228>.

We are grateful to the original authors who published the data and materials associated with this project in an OSF repository (Morin & Marttinen Larsson 2024). The repository contains

a “conversation transcript”, from which the iterative training process could be traced. It also contains several datasets for training and testing an LLM to differentiate between evaluative and non-evaluative uses of the verb *consider*, which can be illustrated with the following examples:

- a. Cain is *considered* a master of American hard boiled crime fiction (...) (Morin & Marttinen Larsson 2025: 2)
- b. Please *consider* all classes when punching number in to spread-sheet and designing new items (Morin & Marttinen Larsson 2025: 2).

In example a. the verb *consider* constitutes an evaluation of the subject. Such evaluative constructions can be and are often formulated using *consider as* or *consider to be*. In example b., however, the meaning is equivalent to ‘take into account’ or ‘think about’, and has no evaluative component.

The model outputs classified sentences as ‘evaluative’, ‘non-evaluative’ or ‘?’ in ambiguous cases. They were evaluated against a human gold standard also provided in Morin & Marttinen Larsson (2024).

In this study, we seek answers to the following three research questions (RQs):

1. Is the “replicable method” proposed in Morin & Marttinen Larsson (2025) independently replicable using the information and materials made available by the authors in the publication and associated OSF repository?
2. Can smaller, open-source models perform as well or better on Morin & Marttinen Larsson (2025)’s *consider* annotation task as Claude Sonnet 3.5 based on the accuracy metrics reported in (Morin & Marttinen Larsson 2025: 6)?
3. If RQ1 is answered in the positive, can this also be achieved with a significantly less extensive training process than that proposed in Morin & Marttinen Larsson (2025)?

While the close replication seeks answers to RQ1 and RQ2, our extended replication serves to answer RQ3. We test the performance of two open-source LLMs: Mistral Small 3.2 24B Instruct 2506 (MistralAI 2026) (hereafter referred to as ‘Mistral 24B’), an open-weight model released in 2025 under Apache 2.0 license by the French AI company Mistral AI, and OpenAI GPT OSS 120B (OpenAI 2026), an open-weight large language model that was also released under Apache 2.0 license by OpenAI (hereafter ‘GPT 120B’).

The outcome of the replications will be compared and assessed based on the metrics Morin & Marttinen Larsson (2025) provided in their evaluation: Accuracy, Precision, Recall, F1 score, and Matthew’s correlation coefficient (Morin & Marttinen Larsson 2025: 6). We formulate the following hypotheses:

Hypothesis 1: The model OpenAI GPT OSS 120B performs as well as or better than Claude Sonnet 3.5 on the *consider* task, as measured by the accuracy metrics reported in Morin & Marttinen Larsson (2025).

Hypothesis 2: The model Mistral Small 3.2 24B Instruct 2506 performs as well as or better than Claude Sonnet 3.5 on the *consider* task, as measured by the accuracy metrics reported in Morin & Marttinen Larsson (2025).

Hypothesis 3: The annotation task can be performed with a significantly less extensive training process than that used by Morin & Marttinen Larsson (2025), while achieving performance comparable to or better than Claude Sonnet 3.5 according to the accuracy metrics reported in Morin & Marttinen Larsson (2025).

3 Close replication

Morin & Marttinen Larsson (2025) present a methodological pipeline leveraging an LLM for the automatic annotation of grammatical data, based on the case study of the evaluative *consider* construction in English. Having engaged in a complex “conversation” with Claude 3.5 Sonnet over several iterative phases of training and testing (see Figure 2), Morin & Larsson (2026): 6] were able to achieve an accuracy of 93% compared to the gold standard provided by a human annotator.

Table 1 compares the original accuracy metrics with those of our close replication in which we attempted to follow the “replicable method” as closely as possible, using the same data and matching the training and testing to a conversation transcript the authors provided (Morin & Marttinen Larsson 2024), but with two open LLMs. As explained in Section 1, the commercial LLM used in the original study, Claude Sonnet 3.5, was already deprecated by October 2025, just six months after the online publication of the study. Whilst the authors could not have predicted that Anthropic would retire this particular model so quickly, this extremely quick turnaround illustrates the risks of attempting to build reproducible and sustainable research pipelines on the commercial services of AI companies.

In answer to RQ1, when designing our close replication, we found that essential information to independently replicate the annotation task was missing, forcing us to make some educated guesses as to the exact procedure. Most importantly, the original paper does not disclose how the pipeline was implemented at the programmatic level. This includes whether the web interface or an API was used¹, how the model parameters were configured, which programming language was used, and how the datasets were provided as input to the models. Since the article only provides an outline of the iterative training approach, many details of the actual

¹While this information was missing in Morin & Marttinen Larsson (2025), the authors disclosed that they used Anthropic’s web chat interface in a more recent preprint (Morin & Larsson 2026). We designed and conducted our replication studies prior to this publication and were not aware of this follow-up study until recently.

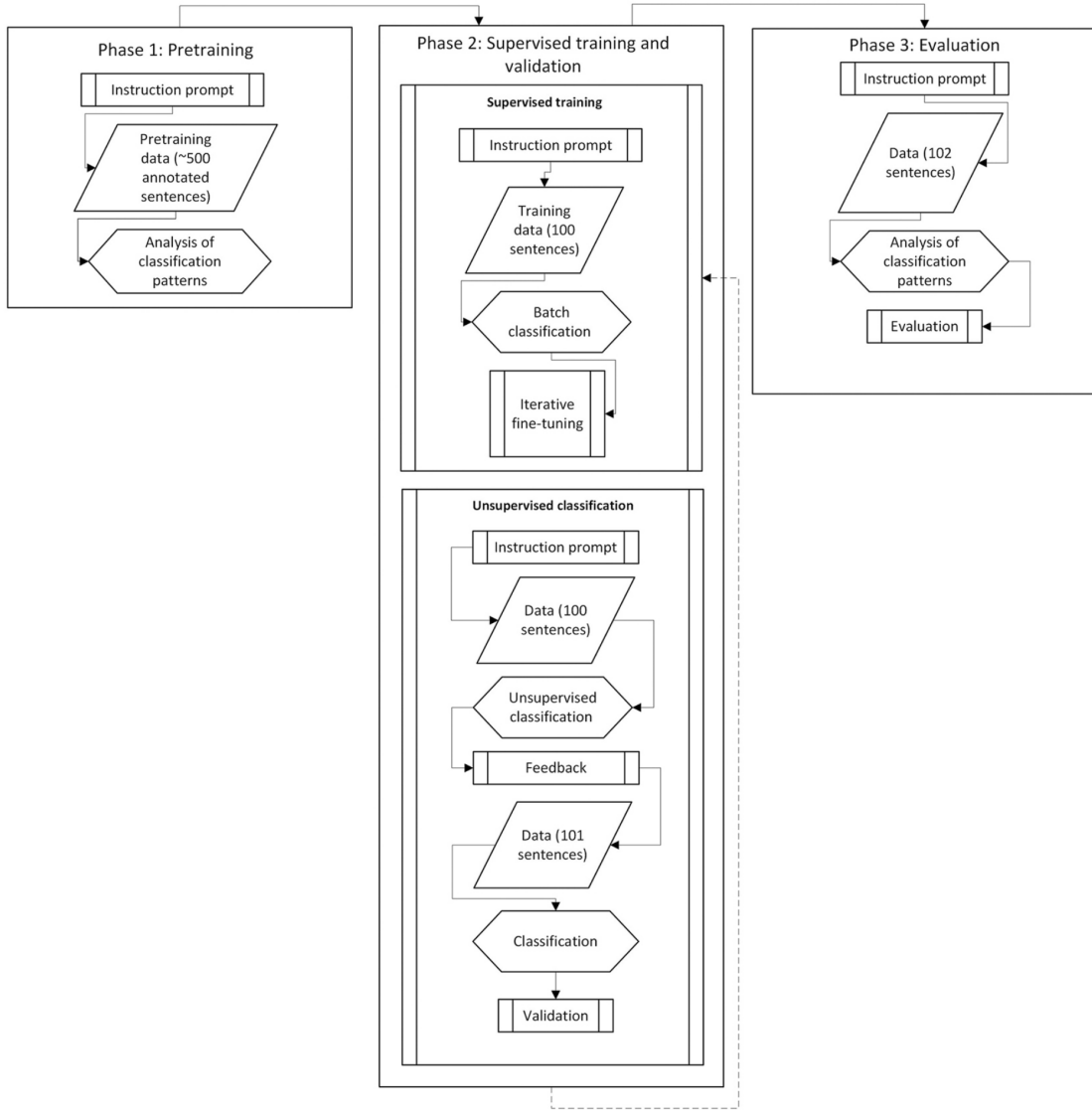


Figure 2: Morin & Marttinen Larsson (2025)’s pipeline for LLM-assisted annotation of linguistic data, based on the case study of consider constructions (CC BY Morin & Marttinen Larsson 2025: 5)

process had to be inferred from the so-called “conversation transcript” provided in the OSF repository, which contains all the “answers” and explanations given to the model during the training phases. It was unclear to us whether the authors used Claude’s web interface or an API when we designed our replication studies, but we chose to use an API and to share all of our scripts to ensure that our replications are as reproducible as possible (see Data Availability Statement). Whilst not essential, having access to the authors’ original analysis code would also have been useful to ensure more precise comparability of our results.

Table 1: Classification metrics of the three models compared to the human gold standard

Metric	Mistral 3.2 24B	GPT OSS	
		120B	Claude 3.5 Sonnet
Accuracy	67%	94%	93%
Precision	77.8%	90.2%	88.1%
Recall	17.9%	94.9%	94.9%
F1-score	29.2%	92.5%	91.4%
Matthews correlation coefficient	0.25	0.88	0.86

Based on accuracy compared to the original human gold standard, our open 120B GPT model showed comparable and even slightly better performance than Claude in the original study (GPT: 94% vs. Claude 93%), while 24B Mistral performed considerably worse (67%). This observation is reflected in the other metrics as well: While the GPT model’s performance is similar or better in all evaluated metrics, Mistral’s much smaller model fails to achieve comparable results. While Mistral’s 67% accuracy might initially sound moderately good, investigating these results as 2 x 2 matrices (see [0_Appendix.pdf](#)) reveals that it is heavily biased toward predicting ‘non-evaluative’ meaning. This poor accuracy is best observed in the extremely low recall of 17.9%. Given the distribution of *consider*-meanings in the testing dataset (‘non-evaluative’ was the majority class accounting for 63/102, 62% of examples), a classifier that always predicted the majority class would also have achieved an accuracy of 62%. The imbalance in this dataset is common to many linguistic annotation tasks and serves as a reminder that, as did Morin & Marttinen Larsson (2025)[Table 1], it is crucial to consider a range of metrics – not just accuracy – when assessing model performance.

The results suggest that the (close) replicability of the approach depends on the comparability of the models. Recalling RQ2, the open GPT model performed slightly better than Claude 3.5 Sonnet, whereas the smaller Mistral model did not. We therefore have strong evidence in support of hypothesis 1, but can reject hypothesis 2. On a methodological level, we found that essential information to independently replicate the study was missing, forcing us to make some educated guesses as to the exact procedure. Most importantly, the paper does not disclose how the pipeline was implemented at the programmatic level. This includes e.g. whether the web interface or an API was used, how the model parameters were configured, which programming language was used, and how the datasets were provided as input to the models. Access to the

analysis code would have been very useful to ensure a close replication of the original approach. Since the paper only outlines the iterative training approach, many details of the actual process could only be inferred from the ‘conversation transcript’, which also contains all the ‘answers’ and explanations given to the model in unsupervised training.

4 Extended replication

Our close replication raised questions about the need for large commercial LLMs to complete this kind type of binary linguistic classification task. With this extended replication, we turn to testing the need for the extensive ‘in-conversation’ training and evaluation process outlined in Morin & Marttinen Larsson (2025). To this end, we applied a much simpler few-shot training approach with the best-performing open LLM from our first replication using the same data, GPT OSS 120B.

As shown in Table 2, when instructing the GPT model only using the pretraining set and then directly proceeding to testing, GPT achieved an accuracy of 94% (96/102). While the accuracy is equivalent to the GPT result in the close replication, the other metrics reveal some minor differences in classification performance: Recall is 100%, meaning that GPT did not miss any positives (‘evaluative’ sentences) in the extended replication. Compared to the original study, the accuracy metrics are comparable: Precision is slightly lower (GPT: 86.7% vs. Claude: 88.1%), meaning that the open GPT model predicted marginally more false positives (‘evaluative’ for actual ‘non-evaluative’). All the other metrics are higher than then the original Claude model. We conclude that these results provide support for our second hypothesis.

Table 2: Classification metrics of both GPT analyses and Claude compared to the human gold standard

Metric	GPT (close replication)	GPT (reduced training)	Claude (original study)
Accuracy	94%	94%	93%
Precision	90.2%	86.7%	88.1%
Recall	94.9%	100%	94.9%
F1-score	92.5%	92.9%	91.4%
Matthews correlation coefficient	0.88	0.89	0.86

5 Discussion

Our results suggest that, when using a comparable open-source LLM, similarly high accuracy rates can be accomplished with a less elaborate training process as in Morin & Marttinen Larsson (2025). However, the smaller Mistral model showed a strong tendency toward the majority class and performed only slightly above the majority-class baseline, indicating to not be suitable to successfully complete this kind of linguistic task.

In a recent paper following up on this project, Morin & Larsson (2026) themselves propose a much simpler workflow. At the same time, their focus seems to distance itself from the goal of an easily accessible and replicable pipeline, as it now involves some programming using the API, and processing the data in batches (Morin & Larsson 2026).

5.1 Suitability of LLMs and training process

We consider this replication successful with regard to the replicability of the methodological pipeline using an open LLM (OpenAI GPT OSS 120B), both with the iterative training outlined in Morin & Marttinen Larsson (2025) and with a few-shot prompting approach. However, the smaller Mistral model failed to achieve the desired accuracy metrics, suggesting that the approach may only be replicable with models comparable in size and functionality.

As the results of this replication indicate, an open-source LLM can achieve performance comparable to that of a closed LLM on the linguistic classification task in Morin & Marttinen Larsson (2025). As already mentioned in the introduction, open models constitute an integral part of open research, which is why we consider this aspect very relevant when doing research with LLMs. In addition to their limited availability, closed, commercial LLMs are often developed intransparently and without ethical oversight (Liesenfeld, Lopez & Dingemanse 2023). Since their training relies on vast amounts of data, including text from the internet, publicly available datasets, or proprietary sources, they are susceptible to passive privacy leakage, whereby sensitive information contained in training data may be revealed. Also, LLMs can be vulnerable to different kinds of active privacy attacks (Yan et al. 2025: 1). However, these risks affect open LLMs as well.

5.2 Replicable methods

This replication contributes to the broader discussion of replicability in linguistic research involving LLMs by examining not only whether the methodological pipeline of the original study can be implemented with different (open) models and amounts of training, but also which aspects of the pipeline and paper allow or constrain reproducibility and replicability. This is particularly important for method-focused studies using commercial LLMs, where access to the model and its underlying configurations are limited. Rather than seeing replicability

as a binary outcome, our results illustrate that different components of such a paper may be replicable to different degrees.

In view of this, several aspects of the original study limit the extent to which the results can be replicated independently. Some important details were not reported by Morin & Marttinen Larsson (2025), as for example the information that they used a chat interface instead of an API (which is later mentioned in Morin & Larsson (2026)) or further details on the implementation were not disclosed. In addition, the human gold standard used for evaluation seems to be established by a single annotator. It is not clear if inter-rater agreement was assessed during its creation. Such limitations leave some uncertainty about whether differences between the original and replication results are attributable to the models, implementation details, or other aspects.

Another thing that we want to address in terms of replicability is the claim that this pipeline may be applied to other tasks and other languages. While the authors already stress that the exact conditions will vary based on the task and some languages will be harder to study due to less available training data (Morin & Marttinen Larsson 2025: 7), existing research has predominantly focused on English-language contexts and paid less attention to LLMs trained on or evaluated in other languages (Yao et al. 2025: 2; Kargaran et al. 2025). Studies focusing on reasoning capabilities of LLMs show multilingual performance to be uneven on the explicit reasoning level, as models trained on English-centric corpora often struggle with underrepresented languages due to limited multilingual reasoning training data, weaker language understanding ability, and lower-quality reasoning. Latent reasoning processes are still almost exclusively examined on English (Liu et al. 2026: 22330). Open-source models offer another advantage for linguistic research in that regard, as in some cases, pretraining data and language distribution are made transparent.

5.3 Ethical and environmental concerns

We also believe that smaller, locally-run LLMs are worth considering for environmental reasons: Training and operating LLMs with billions of parameters requires substantial computational resources, including large amounts of water (e.g. for cooling data centers) and electricity, generating substantial carbon emissions (Özcan et al. 2025; Kron 2026). Moreover, the environmental impact of LLM use depends not only on the model itself but also on the complexity and length of individual prompts, with more demanding prompts generally requiring more computational resources than shorter and less complex ones (Jegham et al. 2025: 11-12). Although AI becomes cheaper and faster, its global adoption is rapidly increasing resource consumption (Jegham et al. 2025).

6 Conclusion

This study investigated the replicability of an LLM-based methodological pipeline for linguistic data annotation. While the smaller model did not achieve comparable results, Morin & Marttinen Larsson (2025)’s approach was successfully replicated with one open model, and also with a significantly less extensive training approach. At the same time, it is worth noting that the study by Morin & Marttinen Larsson (2025) represents an early contribution to this specific area. Regarding the generalisability of this method, it is important to consider that the grammatical construction annotated here constitutes a rather clear-cut formal case (Morin & Marttinen Larsson 2025: 7). Even with promising results from some LLMs, further exploration is needed to effectively and reliably leverage them for linguistic data annotation, especially when they are used for more complex tasks or under-researched languages.

The extensions explored in this study give insights into what can be learned from testing LLM-based methods beyond the conditions of the original study. They illustrate the importance of investigating not only which results an approach produces, but also under which conditions it remains reliable and replicable. As LLMs are increasingly used in corpus-based research, open-science practices, such as making datasets, code, model parameters, and evaluation procedures as transparent as possible, become particularly relevant.

More broadly, this study highlights the importance of replicability in linguistic research with LLMs and the advantages of using open models, which can offer at least some amount of control, and greater possibilities for reproducibility. To leverage LLMs in linguistic research, it is important to weigh their potential and their limitations, rather than assuming they can be applied to every task reliably. This also means considering broader issues associated with their usage, ranging from data rights to environmental impact.

Data availability statement

The data and code that support the findings of this study are openly available in [repository name e.g “figshare”] at [http://doi.org/\[doi\]](http://doi.org/[doi]), reference number [reference number]. The current repository has been anonymised for double-blind peer review, but will be de-anonymised on acceptance.

CRediT author statement

Author1: Methodology, Data Curation, Code, Validation, Formal analysis, Investigation, Writing - Original Draft, Writing - Review & Editing, Project administration. **Author2:** Conceptualisation, Methodology, Validation, Investigation, Writing - Original Draft, Writing - Review & Editing, Supervision, Project administration.

Declaration of conflict of interest

The authors have no conflicts of interest to declare.

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